

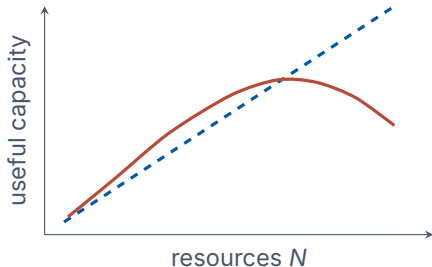


Scalable Distributed Computing - 2026/27

Lectures 3 - Sources and Limits of Scalability I

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What are these three lessons trying to explain?

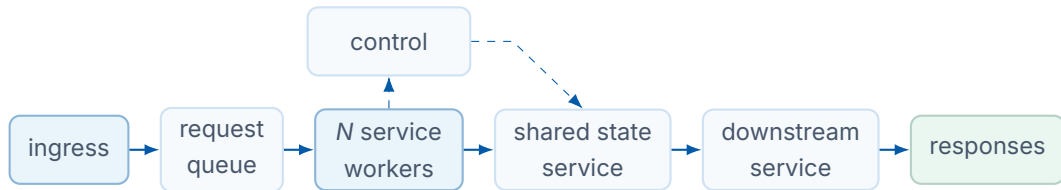


The ideal curve is easy to write. The real intellectual work begins when the measured curve departs from it.

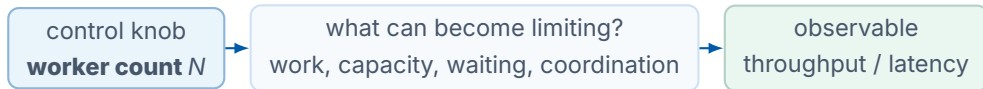
- identify a plausible mechanism;
- build the smallest useful quantitative model;
- derive a prediction that could distinguish that mechanism from alternatives.

Poor scalability is an observation, not a diagnosis.

A reference system we will keep returning to

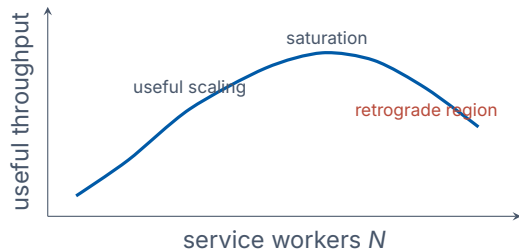


This is not one specific product. It is a deliberately small reference architecture for a common scalable service: requests enter, workers process them, every request touches shared state, and the path continues through a downstream service.



The same boxes and arrows will reappear. Each model will highlight a different reason why this end-to-end path can stop scaling.

A recurring anomaly: more workers, less useful throughput



Nothing in this path says which component is responsible. The same outward symptom can come from very different internal mechanisms.

Our task is not to name the curve. It is to explain why this system could produce it.

Why does scaling bend?

This is not what the ideal model predicted

$$T_{\text{ideal}}(N) = \frac{W}{N}$$



$$S_{\text{ideal}}(N) = N$$

If resources double, ideal capacity doubles or ideal completion time halves.

The observed system did something qualitatively different: **it reached a peak and then became worse.**

The scalability gap

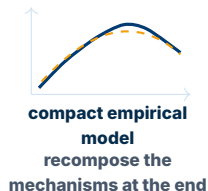
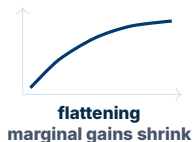
$$T_{\text{real}}(N) > T_{\text{ideal}}(N) = \frac{W}{N}$$

$$G(N) = T_{\text{real}}(N) - T_{\text{ideal}}(N)$$

Where does $G(N)$ come from?

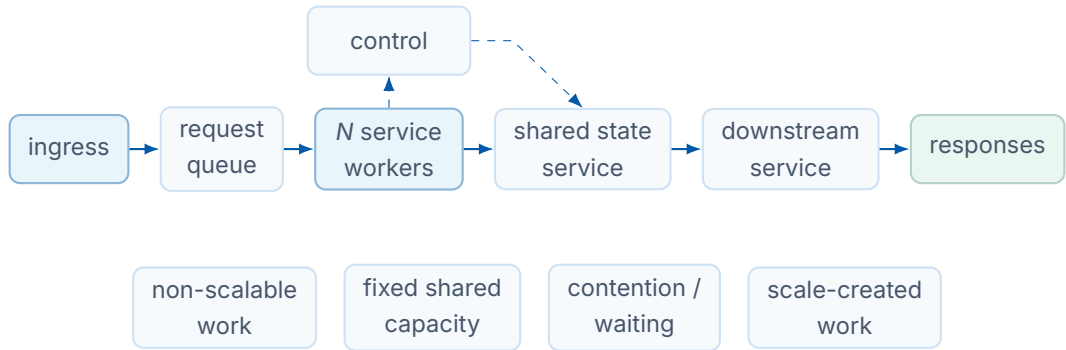
The curve shapes we are trying to explain

Before we derive anything, keep the destination visible. Different mechanisms leave different qualitative signatures in a scaling curve.



We will first understand the mechanisms separately. Only near the end will we ask whether one compact law can summarize several of these regimes.

The first discipline: keep competing explanations alive



Same symptom, different mechanisms, different predictions.

Why Amdahl again?

Two laws, two questions

Amdahl

fixed work

How much faster?

diagnostic baseline

Gustafson

scaled useful work

How much more work?

scale-out objective

Different questions, not competing laws.

Gustafson is close to weak-scaling intuition, without being identical to every formal definition of weak scalability.

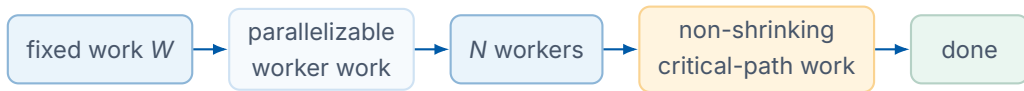
ACT I

Non-scalable work

Amdahl as the simplest diagnostic baseline.

First diagnostic experiment: freeze the workload

Amdahl answers a deliberately narrower question than the production service normally poses. To isolate one mechanism, imagine a **fixed amount of useful work** passing through the same architecture while N changes.



This is an analytical probe, not a claim that the online service has become a batch program. We freeze the workload only to ask whether a non-scalable fraction alone could explain the bending.

Use the scalability law we already have

For the fixed-workload experiment, let

$$s = \frac{T_{\text{serial}}}{T(1)}, \quad 1 - s = \frac{T_{\text{parallel}}}{T(1)}.$$

Amdahl gives

$$S(N) = \frac{1}{s + \frac{1-s}{N}}.$$

We already know the law. Now we use it to derive **marginal gain**, **concavity**, and **resource requirements**.

Amdahl, AFIPS 1967.

Put the known law in a form suited to today's question

Start from

$$S(N) = \frac{1}{s + \frac{1-s}{N}}.$$

Multiply numerator and denominator by N :

$$S(N) = \frac{N}{sN + (1 - s)}.$$

Therefore

$$S(N) = \frac{N}{sN + 1 - s}.$$

Same law, different form. This form makes derivatives and sensitivity much easier to inspect.

The known asymptotic result is not enough

From the earlier scalability-law analysis,

$$\lim_{N \rightarrow \infty} S(N) = \frac{1}{s}.$$

For $s = 0.05$,

$$S_{\max} = 20.$$

That establishes the ceiling, but it does not yet answer two engineering questions:

How fast do marginal gains disappear?

How much scale is needed to get close?

What more can we deduce from the same law?

A deliberate continuous relaxation

Machines are discrete:

$$N \in \mathbb{N}.$$

For shape analysis we temporarily extend the model to

$$N \in \mathbb{R}^+.$$

This lets us use derivatives to study marginal gain and curvature. The result is an analytical approximation to the discrete deployment problem, not a claim that fractional machines exist.

Modelling note. The continuous relaxation is an analytical tool. For discrete decisions we will either use finite differences when they have a direct marginal interpretation, or compare the neighboring integer configurations around a continuous optimum.

What is the marginal value of more resources?

Start from

$$S(N) = \frac{N}{sN + 1 - s}.$$

Differentiate:

$$S'(N) = \frac{(sN + 1 - s) - sN}{(sN + 1 - s)^2}.$$

Therefore

$$S'(N) = \frac{1 - s}{(sN + 1 - s)^2}.$$

$S'(N)$ is the local marginal speedup predicted by the continuous relaxation.

The gain is positive - but how fast does it disappear?

Because

$$S'(N) = \frac{1 - s}{(sN + 1 - s)^2},$$

the numerator is constant while the denominator grows with N . Therefore

$$\boxed{\lim_{N \rightarrow \infty} S'(N) = 0}.$$

The marginal benefit of additional resources tends to zero.

The asymptote tells us **where** the system is heading. The first derivative tells us **how much value the next increment of scale still has**.

One serial fraction, very different marginal returns

Take the same model with $s = 0.05$ and evaluate it at increasingly large resource counts.

N	$S(N)$	efficiency $S(N)/N$	next-node gain $\Delta S(N)$
4	3.48	0.87	0.63
16	9.14	0.57	0.31
64	15.42	0.24	0.06

The law still predicts improvement at every row, but the economic meaning of “one more resource” changes dramatically.

The important feature is not only the asymptotic ceiling: the **marginal return collapses long before the ceiling is reached.**

Are the marginal gains themselves decreasing?

The first derivative is the marginal gain. To test whether that gain decreases as N grows, differentiate it once more:

$$S'(N) = (1 - s)(sN + 1 - s)^{-2}.$$

Using the chain rule,

$$S''(N) = (1 - s)(-2)(sN + 1 - s)^{-3} \cdot s.$$

Hence

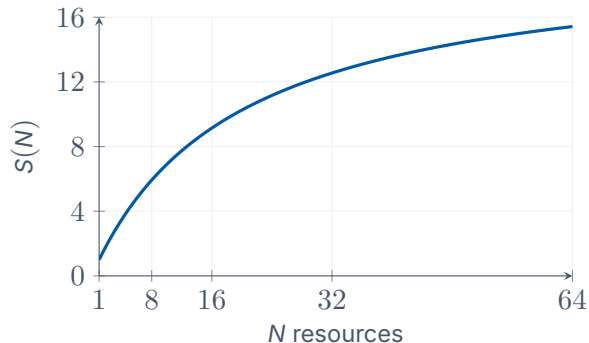
$$S''(N) = -\frac{2s(1 - s)}{(sN + 1 - s)^3}.$$

For $0 < s < 1$ and $N > 0$,

$$S''(N) < 0.$$

The second derivative is not an extra formula for its own sake: it tells us that the marginal benefit quantified by $S'(N)$ is itself shrinking.

Concavity has a system meaning



$$S''(N) < 0$$

means the curve bends downward.

System interpretation

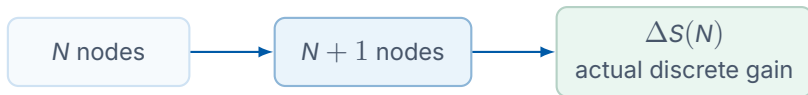
Each additional resource contributes less speedup than the previous one.

Diminishing returns are not a slogan here; they follow from the model.

Machines still come in integers

For a real deployment, the directly interpretable marginal gain is

$$\Delta S(N) = S(N + 1) - S(N).$$



Modelling note. Derivatives expose the structure of the curve; finite differences answer the literal “one more node” question.

How many resources are needed to get close to the limit?

The asymptotic maximum is

$$S_{\max} = \frac{1}{s}.$$

Suppose we want at least a fraction ρ of that maximum:

$$S(N) \geq \rho S_{\max}.$$

Substitute both expressions:

$$\frac{N}{sN + 1 - s} \geq \frac{\rho}{s}.$$

Can we solve this directly for N ?

Solving the sizing inequality

Starting from

$$\frac{N}{sN + 1 - s} \geq \frac{\rho}{s},$$

all denominators are positive for $0 < s < 1$, $N > 0$, so cross-multiplication preserves the inequality:

$$sN \geq \rho(sN + 1 - s).$$

Expand:

$$sN \geq \rho sN + \rho(1 - s).$$

Collect the terms in N :

$$sN(1 - \rho) \geq \rho(1 - s).$$

Therefore

$$\boxed{N \geq \frac{\rho(1 - s)}{s(1 - \rho)}}.$$

The last few percent are expensive

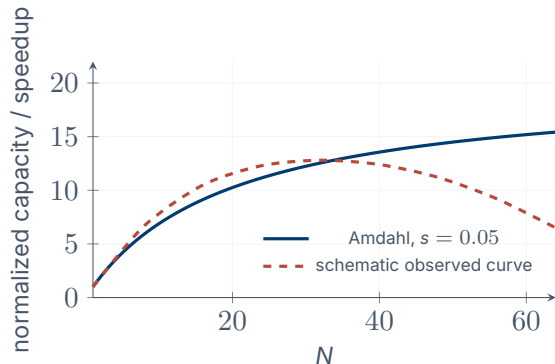
Let $s = 0.05$, so $S_{\max} = 20$.

Target fraction ρ	Target speedup	Minimum continuous N
0.80	16	76
0.90	18	171
0.95	19	361

“The limit is 20” hides an engineering fact: getting very close to 20 can require hundreds of resources.

For a real deployment, round up to an admissible integer resource count. Whether the model describes the real platform is an empirical question; Lecture 4 will develop how to test such claims correctly.

Amdahl can flatten the curve -- but it cannot make it turn down



For $0 < s < 1$,

$$S'(N) > 0.$$

So Amdahl predicts *ever smaller gains*,
but never negative gains.

The opening anomaly therefore
requires another mechanism.

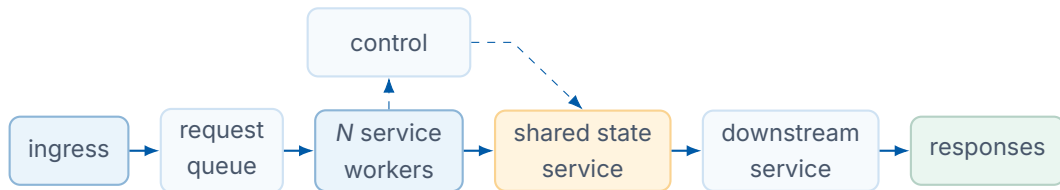
**Can Amdahl explain the
drop?**

ACT II

Fixed shared capacity

The scalable tier is not the whole system.

Now hold one shared-state capacity fixed



The worker tier scales with N . The shared-state service does not automatically gain capacity just because more workers exist.

Every successful request must still cross the highlighted component.

This is the same reference system; only the active hypothesis has changed.

Generalizing the service into a capacity model

Let each worker provide capacity

x requests/s,

so N workers provide

$$X_{\text{workers}}(N) = Nx.$$

Let the required shared resource provide at most

C requests/s.

The request must pass through both stages, therefore end-to-end capacity is

$$X(N) = \min\{Nx, C\}.$$

**The scalable tier and the fixed-capacity tier form one end-to-end system;
the active constraint is the smaller capacity.**

Where does saturation begin?

There are two regimes:

$$Nx < C \Rightarrow X(N) = Nx,$$

$$Nx \geq C \Rightarrow X(N) = C.$$

The transition occurs when

$$Nx = C.$$

Solving for the continuous crossing point gives

$$N^* = \frac{C}{x}.$$

Because worker count is discrete, the first deployable count that can saturate the shared resource is

$$N_{\text{sat}} = \left\lceil \frac{C}{x} \right\rceil.$$

Back to the reference service: how many workers are useful?

Use illustrative capacities:

$$x = 700 \text{ req/s per worker}, \quad C_{\text{state}} = 10,000 \text{ req/s}.$$

Then

$$\frac{C_{\text{state}}}{x} = \frac{10,000}{700} \approx 14.29,$$

so the first integer worker count that can saturate the shared state is

$$N_{\text{sat}} = 15.$$

Prediction if the model is appropriate: Beyond about 15 workers, throughput should remain near the shared-state capacity unless another component becomes limiting first; increasing shared-state capacity should move the plateau.

The model compares two interventions before deployment

At 15 workers the worker tier could offer

$$15 \times 700 = 10,500 \text{ req/s,}$$

so the shared-state service at 10,000 req/s is already the active constraint.

Add 5 workers

$20 \times 700 = 14,000$ req/s worker capacity, but end-to-end throughput remains near 10,000 req/s.

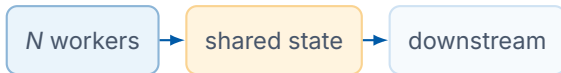
Scale shared state

Raise shared-state capacity to 20,000 req/s: with the same 20 workers, the worker tier can now become the active limit near 14,000 req/s.

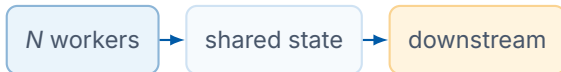
A capacity model is useful because it predicts which intervention can move the plateau.

The same system can move from one bottleneck to another

Before scaling shared state



After scaling shared state



Removing one active constraint does not make the path unlimited; it can simply expose the next capacity limit.

A concrete bottleneck-migration calculation

Keep the same worker capacity $x = 700$ req/s and suppose the path has two shared limits:

$$C_{\text{state}} = 10,000 \text{ req/s}, \quad C_{\text{down}} = 14,000 \text{ req/s}.$$

Before scaling shared state,

$$X(N) = \min\{700N, 10,000, 14,000\}.$$

After raising C_{state} to 20,000 req/s,

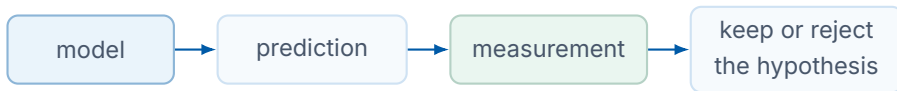
$$X(N) = \min\{700N, 20,000, 14,000\}.$$

The intervention removes the first plateau but predicts a second one near the downstream-service capacity.

Illustrative capacities; the point is the migration of the active constraint along one end-to-end path.

If this explanation is correct, what should we observe?

Prediction if the model is appropriate: As N grows beyond N_{sat} , throughput should remain near C ; the shared resource should approach saturation; extra workers should become increasingly underutilized; increasing C should move the plateau.



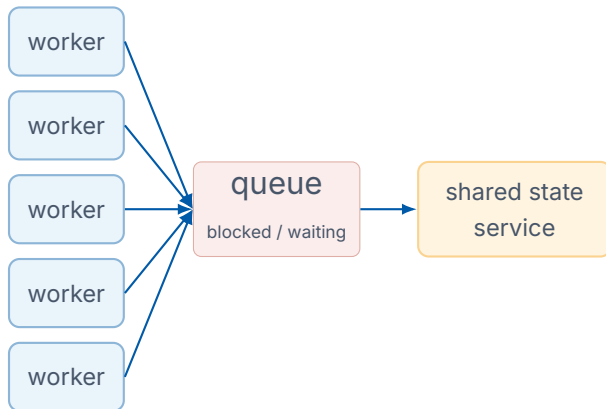
A useful model does not merely fit a curve. It tells us what to measure next.

ACT III

Waiting and scale-created work

Beyond the plateau, the cost of scale can itself grow.

Contention appears as waiting around the same shared state



The bottleneck tells us where capacity stops increasing. Contention tells us what surplus participants do there: they queue, block, retry, or compete for internal serialization points.

Throughput can stay flat while waiting grows

Suppose the shared-state service admits at most 8 concurrent operations and is already continuously busy.

workers	active on shared state	potentially waiting	useful throughput
8	8	0	$\approx C$
16	8	8	$\approx C$
32	8	24	$\approx C$

The reference system has not changed; only the queue in front of the shared-state service has become longer.

A flat throughput curve can hide a growing amount of blocked work.

A minimal contention model and its prediction

A deliberately small decomposition is

$$T(N) = T_{\text{useful}}(N) + T_{\text{wait}}(N).$$

In a regime where contention increases as more workers enter the shared path,

$$\boxed{T'_{\text{wait}}(N) > 0}.$$

Prediction if the model is appropriate: If contention is the dominant explanation, increasing N should produce more queueing, blocking, or lock-wait time around the shared resource, even when useful throughput no longer increases.

Contention and scale-created overhead are close - but not the same

Both can grow with N , and both can eventually produce retrograde scaling. The distinction is **where the lost time comes from**.

Waiting cost

Useful actors cannot proceed because a shared resource or state is busy.

What grows: blocked / queued / stalled time.

Additional work

The larger system actually performs more communication, metadata, synchronization, retry, or coordination operations.

What grows: messages, bytes, metadata operations, synchronization work.

Observe different things: waiting / blocked-time fraction versus total work volume.

The same throughput curve can contain both effects. Keeping them separate gives us stronger discriminating predictions later.

References used in this lecture

- G. M. Amdahl, "Validity of the Single Processor Approach to Achieving Large Scale Computing Capabilities," AFIPS, 1967.
- J. Dean and S. Ghemawat, "MapReduce: Simplified Data Processing on Large Clusters," OSDI, 2004.
- N. J. Gunther, "A General Theory of Computational Scalability Based on Rational Functions," 2008; and *Guerrilla Capacity Planning*.
- B. Schwartz, "Forecasting MySQL Scalability with the Universal Scalability Law," Percona Live, 2011.
- A. B. Bondi, "Characteristics of Scalability and Their Impact on Performance," WOSP, 2000.
(background framing from earlier in the module)

Empirical examples are explicitly separated from guided numerical examples, reference-system diagrams, and deliberately simplified analytical models.



Questions or comments?

L3 - Sources and limits of scalability I

Department of Computer Science - University of Pisa